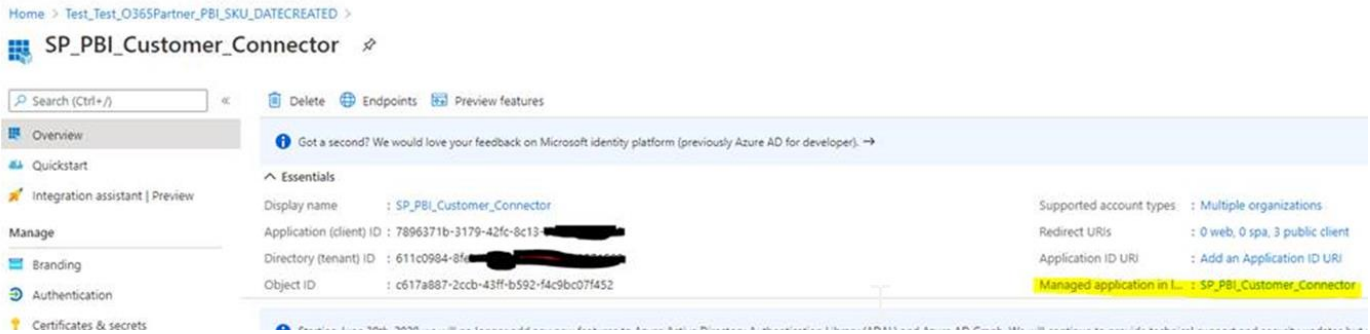
PowerBI sourcing – Authentication procedure

This document explains at a high level, the steps needed to be taken to be able to automate authenticating the PowerBI API service. Health Catalyst TD/DAS team member should closely work with the Azure Administrator of the client to accomplish this.

Register Azure AD application

- Client needs to create a service account for Health Catalyst that can be used for the PowerBI sourcing and add it to a security group
- Register an Azure AD app for this service account in client's Azure environment https://docs.microsoft.com/en-us/power-bi/developer/embedded/embed-service-principal
- When you register the app, at the end you will receive a client(application) ID and client secret – ***Make sure you save the client secret in a secured way. It can be hard to retrieve it later***

Below is a sample app registered (screenshot of Microsoft Support team) and it shows the basic parameters associated with it



- Now, add this registered app to the same security group as service account (step 1 in this section)

Service Principal Authentication

- Service principal (SPN) is an authentication method that can be used to let an Azure AD application access Power BI APIs.
- This mode of authentication has been used to extract PowerBI structure data like workspaces, reports, datasets etc. (non-admin APIs)
- As of February 2021, PowerBI ***admin*** APIs (to get audit data like view, update, create app log) do not support SPN authentication, they will need username/password authentication. This is the direction we got from Microsoft when we worked on this implementation at Honor Health
- However, Microsoft support founded another way to use SPN authentication for both PowerBI structure and admin APIs while we worked with them for Multicare implementation. That is, while you grant permissions to the Azure app (next section of this document), do not provide admin consent (next section – 'Permissions to Azure app' in this document) and that way, you are able to use SPN authentication for all the APIs.

Permissions to the Azure app

- The app should be given all the permissions below to be able to access the API service but DO NOT grant admin consent since this will be handled by the settings on PowerBI portal (explained in next section – Configuration in PowerBI admin portal of this document)
- This approach worked successfully at a recent implementation at Multicare, this should be tried FIRST since we can set up all bindings with the same grant_type

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Applications are authorized to call APIs when they are granted permissions by users/admins as part of the consent process. The list of configured permissions should include all the permissions the application needs. Learn more about permissions and consent

+ Add a permission ✓ Grant admin consent for Multicare Health System

API / Permissions name	Type	Description	Admin consent requ...	Status
Microsoft Graph (1)				
User.Read	Delegated	Sign in and read user profile	No	
Power BI Service (12)				
App.Read.All	Delegated	View all Power BI apps	No	
Capacity.Read.All	Delegated	View all capacities	No	
Dashboard.Read.All	Delegated	View all dashboards	No	
Dataflow.Read.All	Delegated	View all dataflows	No	
Dataset.Read.All	Delegated	View all datasets	No	
Dataset.ReadWrite.All	Delegated	Read and write all datasets	No	
Gateway.Read.All	Delegated	View all gateways	No	
Report.Read.All	Delegated	View all reports	No	
StorageAccount.Read.All	Delegated	View all storage accounts	No	
Tenant.Read.All	Delegated	View all content in tenant	Yes	⚠ Not granted for Multica...
Workspace.Read.All	Delegated	View all workspaces	No	
Workspace.ReadWrite.All	Delegated	Read and write all workspaces	No	

- At Honor (implemented earlier than Multicare), Microsoft support recommended providing Admin Consent after granting the app all the required permissions. But at Honor health, client_credentials grant type did not work for audit/admin APIs, we had to use 'password' grant type

Certificates & secrets
Token configuration
API permissions
Expose an API
App roles | Preview
Owners
Roles and administrators | Preview
Manifest

Support + Troubleshooting
Troubleshooting
New support request

API / Permissions name	Type	Description	Admin consent req...	Status
Azure Active Directory Graph (1)				
User.Read	Delegated	Sign in and read user profile	-	Granted f
Power BI Service (14)				
App.Read.All	Delegated	View all Power BI apps	-	Granted f
Capacity.Read.All	Delegated	View all capacities	-	Granted f
Dashboard.Read.All	Delegated	View all dashboards	-	Granted f
Dataflow.Read.All	Delegated	View all dataflows	-	Granted f
Dataset.Read.All	Delegated	View all datasets	-	Granted f
Gateway.Read.All	Delegated	View all gateways	-	Granted f
Group.Read	Delegated	View user's groups	-	Granted f
Group.Read.All	Delegated	View all groups	-	Granted f
Metadata.View_Any	Delegated	View content properties	-	Granted f
Report.Read.All	Delegated	View all reports	-	Granted f
StorageAccount.Read.All	Delegated	View all storage accounts	-	Granted f
Tenant.Read.All	Delegated	View all content in tenant	Yes	Granted f
Tenant.Read.All	Application	View all content in tenant	Yes	Granted f
Workspace.Read.All	Delegated	View all workspaces	-	Granted f

- The app should be added as an ***admin to all the workspaces*** that the client needs data for. There is a Powershell module available to automate this effort

  - Below link is the Microsoft documentation of this module

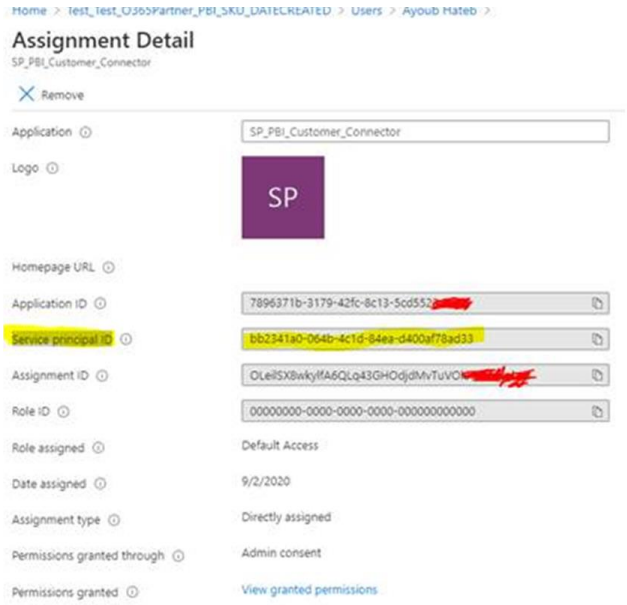
https://docs.microsoft.com/en-us/powershell/module/microsoftpowerbimgmt.workspaces/add-powerbiworkspaceuser?view=powerbi-ps

  - **Command:**

Add-PowerBIWorkspaceUser -Scope Organization -AccessRight Admin -PrincipalType App -Id "XXXXXXX " -Identifier "XXXXXXXX "

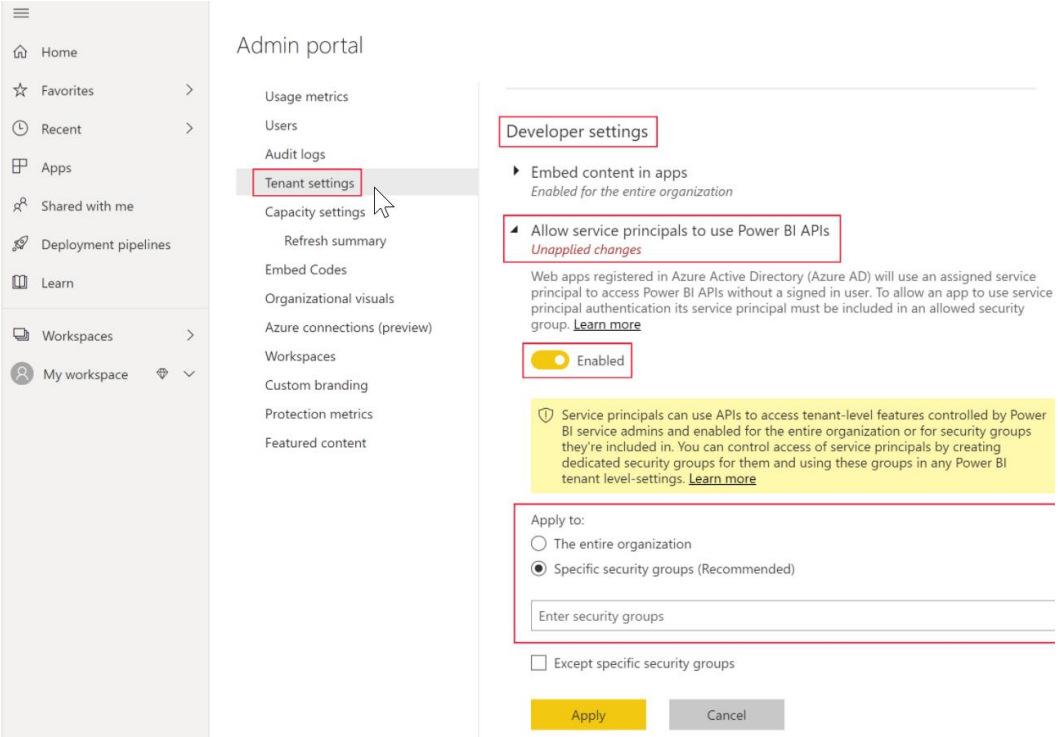
(Where "Id" is the Workspace GUID and the "Identifier" is the GUID of the application user)

Also, the administrator that runs that command has to be an admin of the workspace in order to add a different user.



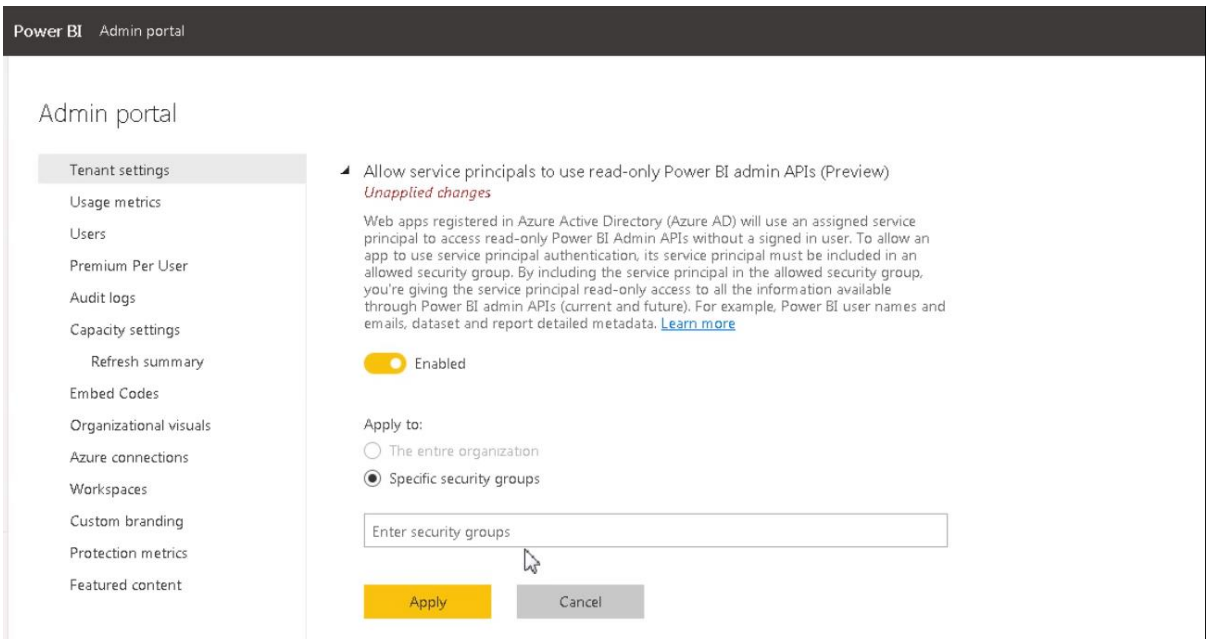
Configuration in PowerBI admin portal

For the app/account to access the APIs, you must enable PowerBI service admin setting. In the 'Apply To' section, add the security group that has the service account and Azure app



Microsoft documentation for this can be found here: https://docs.microsoft.com/en-us/power-bi/developer/embedded/embed-service-principal

For SPN to access admin activity API, enable it in PowerBI admin portal



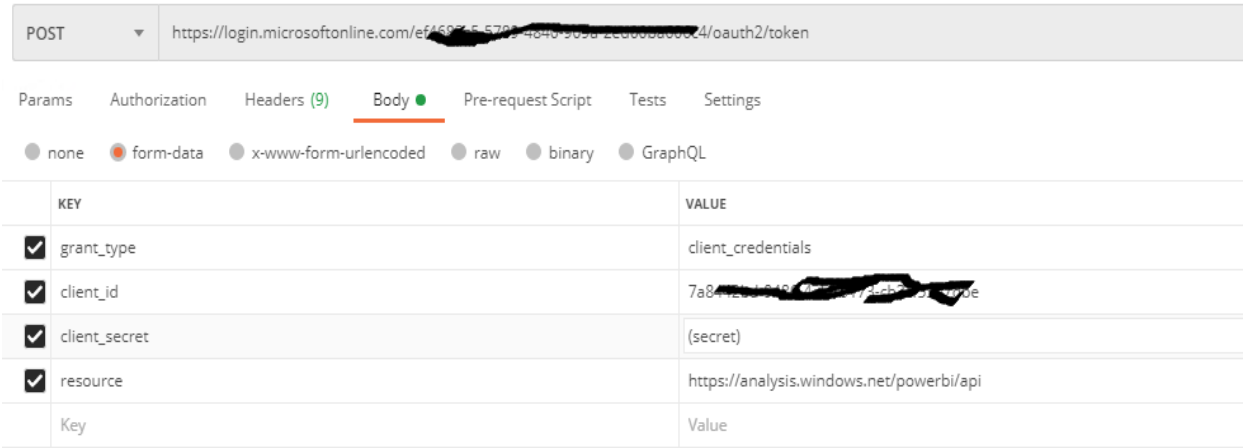
API request using SPN authentication

- At this point, you should be able to make PowerBI REST API calls to get structure data (like workspaces, reports, datasets etc.) by using the Client ID and Client Secret.
- Documentation related to the PowerBI APIs can be found here:

  https://docs.microsoft.com/en-us/rest/api/power-bi/

- Below is a screenshot of making a POST call in postman to get the bearer token

  The end point is standard - https://login.microsoftonline.com/{tenant_ID}/oauth2/token

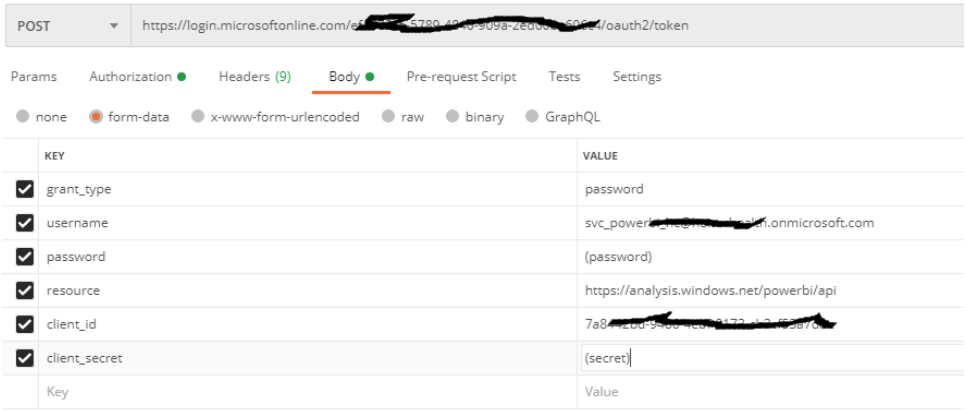


Authenticating Admin APIs using password grant type

- Admin API provide audit/usage data like View log, Create log, Update log etc. Documentation can be found here:

  https://docs.microsoft.com/en-us/rest/api/power-bi/admin/getactivityevents

- As of Feb 2021, SPN authentication is not supported for the PowerBI admin
- For this set of APIs, you'll have to rely on username/password authentication



Additional information for Admin APIs

- If there is MFA setup (Multi Factor Authentication), then the service account will need an MFA exception to use it for API authentication
- If the client's environment is federated, you'll have to get a cloud only account created (.onmicrosoft.com account) – this will be associated with the service account